

JAVA PROGRAMMING LAB RECORD

Experiment No: 05

Date: 21/09/26

Title: List Interface Methods in Java using ArrayList

1. Objective / Aim

To demonstrate commonly used methods of the Java List interface using ArrayList, including adding, accessing, updating, removing, searching, extracting a sub-list, and sorting elements.

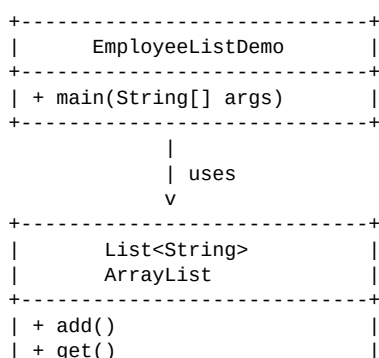
2. System Requirements

Hardware Requirements	Software Requirements
Computer/Laptop	JDK 17 or JDK 21
Minimum 4 GB RAM	Java compiler (javac) and Java runtime (java)
Minimum 1 GHz processor	Eclipse / IntelliJ IDEA / NetBeans / VS Code (optional)
At least 100 MB free storage	Windows / Linux / macOS

3. Algorithm / Logic Description

1. Start the program.
2. Import ArrayList, List, and Comparator from java.util.
3. Create a List using ArrayList.
4. Add employee names using add().
5. Insert an employee at a particular index using add(index, element).
6. Retrieve an element using get(index).
7. Replace an element using set(index, element).
8. Remove an element using remove(index).
9. Find the first and last occurrence using indexOf() and lastIndexOf().
10. Create a portion of the list using subList(fromIndex, toIndex).
11. Sort the list alphabetically using sort(Comparator.naturalOrder()).
12. Display the results and stop.

4. UML Class Diagram



```
| + set() |
| + remove() |
| + indexOf() |
| + lastIndexOf() |
| + subList() |
| + sort() |
+-----+
```

5. Source Code

```
import java.util.ArrayList;
import java.util.List;
import java.util.Comparator;

class EmployeeListDemo {

    public static void main(String[] args) {

        // Creating a List
        List<String> employees = new ArrayList<>();

        // add(E e)
        employees.add("Anjali");
        employees.add("Vikram");
        employees.add("Rohan");
        employees.add("Anjali");

        System.out.println("Original List: " + employees);

        // add(int index, E element)
        employees.add(1, "Meena");
        System.out.println("After add(1, \"Meena\"): " + employees);

        // get(int index)
        System.out.println("Element at index 2: " + employees.get(2));

        // set(int index, E element)
        employees.set(2, "Karthik");
        System.out.println("After set(2, \"Karthik\"): " + employees);

        // remove(int index)
        employees.remove(3);
        System.out.println("After remove(3): " + employees);

        // indexOf(Object o)
        System.out.println("First index of Anjali: "
            + employees.indexOf("Anjali"));

        // lastIndexOf(Object o)
        System.out.println("Last index of Anjali: "
            + employees.lastIndexOf("Anjali"));

        // subList(int fromIndex, int toIndex)
        List<String> selectedEmployees = employees.subList(1, 3);
        System.out.println("SubList from index 1 to 3: "
            + selectedEmployees);

        // sort(Comparator)
        employees.sort(Comparator.naturalOrder());
        System.out.println("After Sorting: " + employees);
    }
}
```

6. Completion and Execution Commands

1. Save the source code as: EmployeeListDemo.java
2. Open Command Prompt / Terminal in the source-code folder.
3. Compile: javac EmployeeListDemo.java
4. Execute: java EmployeeListDemo
5. Verify the output displayed in the terminal.

7. Output

```
Original List: [Anjali, Vikram, Rohan, Anjali]
After add(1, "Meena"): [Anjali, Meena, Vikram, Rohan, Anjali]
Element at index 2: Vikram
After set(2, "Karthik"): [Anjali, Meena, Karthik, Rohan, Anjali]
```

```
After remove(3): [Anjali, Meena, Karthik, Anjali]
First index of Anjali: 0
Last index of Anjali: 3
SubList from index 1 to 3: [Meena, Karthik]
After Sorting: [Anjali, Anjali, Karthik, Meena]
```

8. Result

The EmployeeListDemo program was successfully compiled and executed. The required List interface methods were demonstrated using ArrayList and the expected results were obtained.